

DATABASE SYSTEMS

Chap 1~ Overview of Database Management Systems

1. Overview of Database Management System

- After completing this chapter, the reader should be familiar with the following concepts:
- Data, information, database, database management system
- Purpose of Database Systems
- View of Data
- Data Models
- Database Users
- Overall System Structure



1.1 Data and Information

- ✓ Data are raw facts that constitute building block of information.
- ✓ Data are the heart of the DBMS
- ✓ Useful information is obtained from processed data.
- ✓ Example:
 - **A company needs to save information about employees, departments, and salaries. Those pieces of information are called data.**
- ✓ After processing **DATA** then output data can be perceived as **INFORMATION**.

1.2 Database

- ✓ A database is a well-organized **collection of data** that are related in a meaningful way, which can be accessed in different logical orders.
- ✓ The database should contain all the data needed by the organization.
- ✓ A DATABASE contains information about a particular enterprise.

1.2 Database

✓ Applications of Database.

- **Banking:** all transactions
 - **Airlines:** reservations, schedules
 - **Universities:** registration, grades
 - **Sales:** customers, products, purchases
 - **Manufacturing:** production, inventory, orders, supply chain
 - **Human resources:** employee records, salaries, tax deductions.
- Databases touch all aspects of our lives

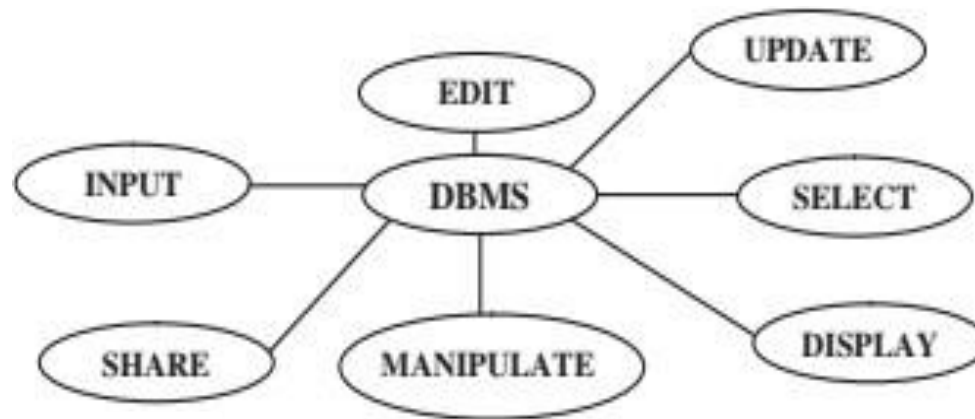
Types of databases

- Number of users:
 - Single-user database: Desktop database
 - Many users : Enterprise database
- Location:
 - Centralized database
 - Distributed database

1.3 Database Management System

✓ A DBMS consists of:

- A set of application programs used to access, update, and manage data. (MySQL, PostgreSQL, and Microsoft SQL Server).
- A DBMS is a general-purpose software i.e., not application specific.
- A DBMS takes care of storing and accessing data.



Capabilities of database management system

1.3 Database Management System

Examples of DBMS Products

- Open source DBMS :

PostgreSQL, MySQL, MariaDB
MongoDB, SQLite.

Structure Dbms

Non-structured Dbms

- Proprietary DBMS: Oracle,

Microsoft SQL Server, IBM
DB2, SAP HANA

Structure Dbms

Non-structured Dbms

1.4 Structured and Unstructured Dbms

- ✓ Structured data follow a strict format , data are organized in Table of row and column. Examples: Customer name, age
- ✓ Example Structured Dbms: PostgreSQL, MySQL
- ✓ Unstructured data has no strict format, data are stored in data lakes, Examples: videos, photos, voice recording
- ✓ Example Structured Dbms: MariaDB, MongoDB

1.4 Purpose of Database System

Main advantages of Database System

✓ Data Availability

- Data are made available to wide variety of users in a meaningful format at reasonable cost

✓ Data Integrity

- Refer to the correctness of the data in the database ~ available data are reliable

✓ Data Security

- Refer to the fact that only authorized users can access the data
- If two separate users are accessing particular data at the same time, DBMS must not allow them to make conflicting changes

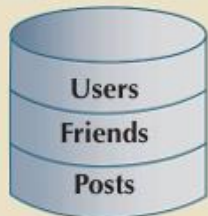


1.4 Database Flexibility for data Accessibility

Before leaving for work, Susan checks her Facebook and Twitter accounts



Where is the data about the friends and groups stored?
Where are the “likes” stored and what would they be used for?



On her lunch break, she picks up her prescription at the pharmacy



Where is the pharmacy inventory data stored?
What data about each product will be in the inventory data?
What data is kept about each customer and where is it stored?



After work, Susan goes to the grocery store



Where is the product data stored?
Is the product quantity in stock updated at checkout?
Does she pay with a credit card?



At night, she plans for a trip and buys airline tickets and hotel reservations online



Where does the online travel website get the airline and hotel data from?
What customer data would be kept by the website?
Where would the customer data be stored?



Then she makes a few online purchases



Where are the product and stock data stored?
Where does the system get the data to generate product “recommendations” to the customer?
Where would credit card information be stored?



1.5 Classification of DBMS

✓ DBMS can be classified into:

- **Passive DBMS**

- client controls DBMS updates,
- program-driven system
- User query the current state of database and retrieve the information currently available in the database.
- The operations can be definitions and updates of the schema, as well as queries and updates of the data.

Example: Library management system

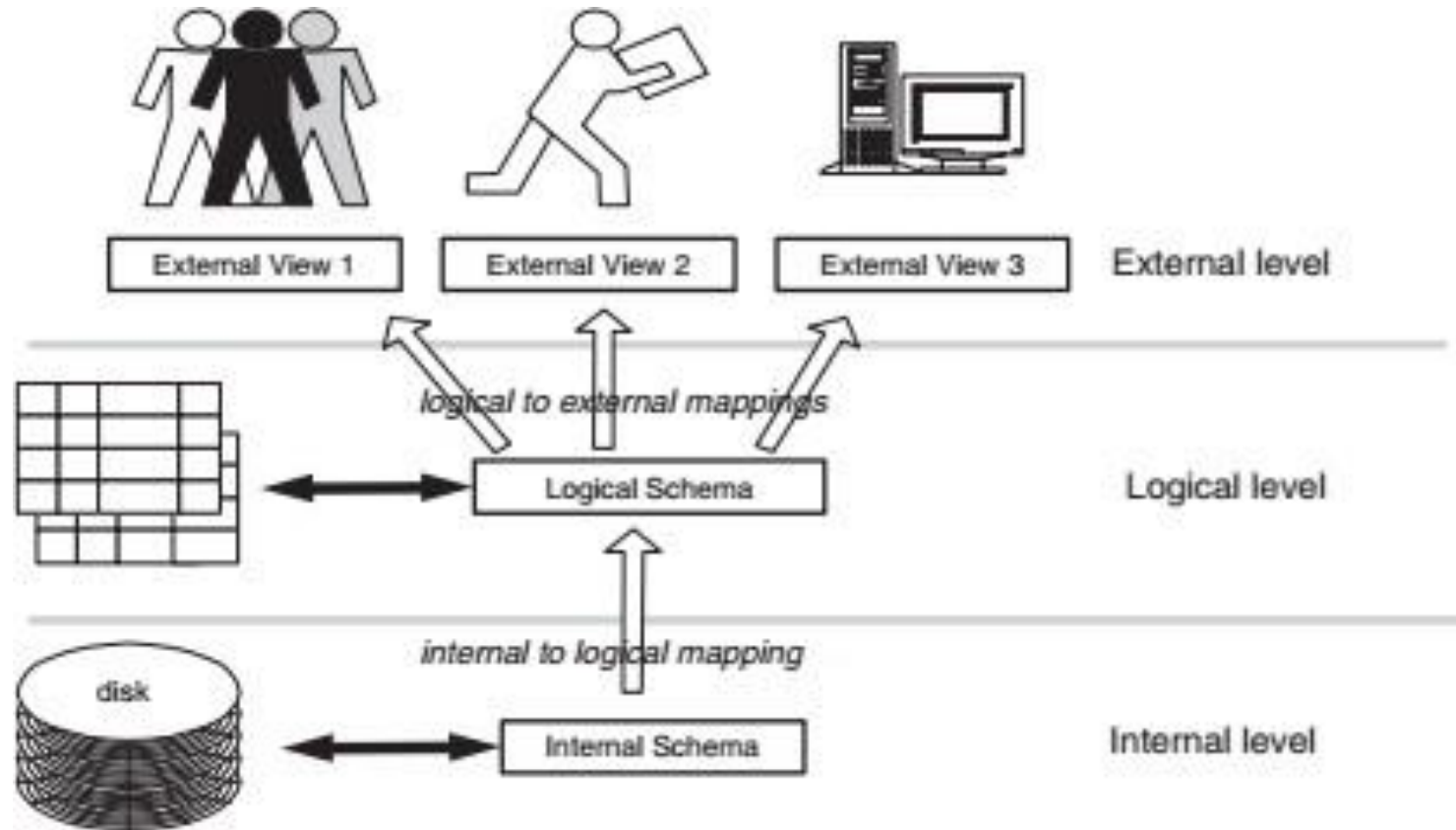
- **Active DBMS**

- An active database system (ADBS) is a DBS that monitors situations of interest and, when they occur, triggers an appropriate response in a timely manner.
- The desired behavior is expressed in production rules (also called event-condition-action rules), which are defined and stored in the DBS
- Example: **Airline reservation system.**

1.6 ANSI/DATA MODEL

- ✓ Data model describe different ways on which data within a database can be viewed.
- ✓ designed to separate the user's view of the data from the physical storage details.
- ✓ ANSI proposed an three-level architecture:
 - **Physical Level or Internal level**: describes the implantation of database(table, indexing) how data (e.g., customer) is stored.
 - **Logical Level or Conceptual level**
 - Describe what data are stored in the database and what relationships exist among those data ,do not rely on any DBMS.
 - **View Level or External level**
 - The external level defines how individual users or applications interact with the database through their customized views

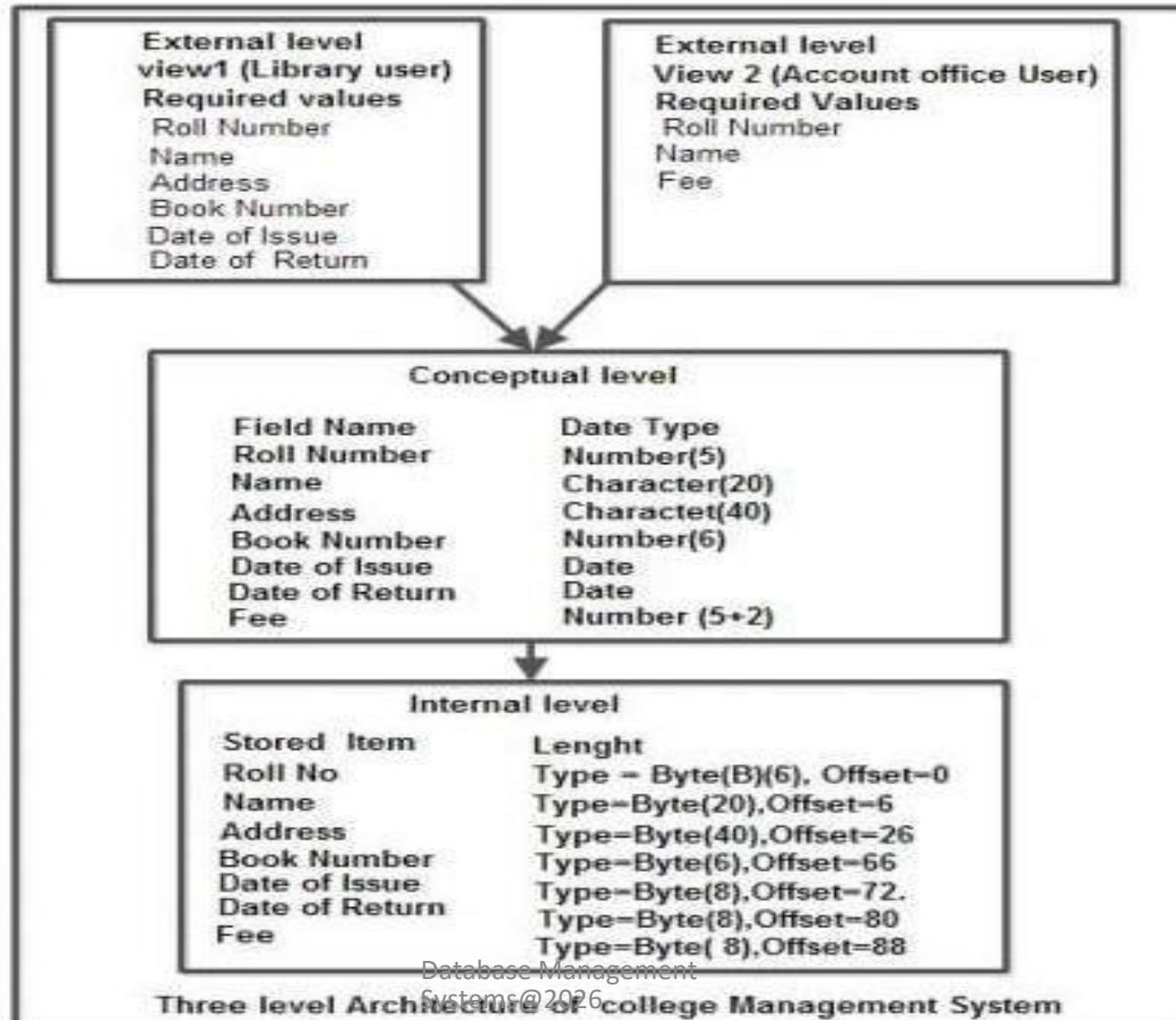
1.6 DATA MODEL



ANSI/SPARK data model

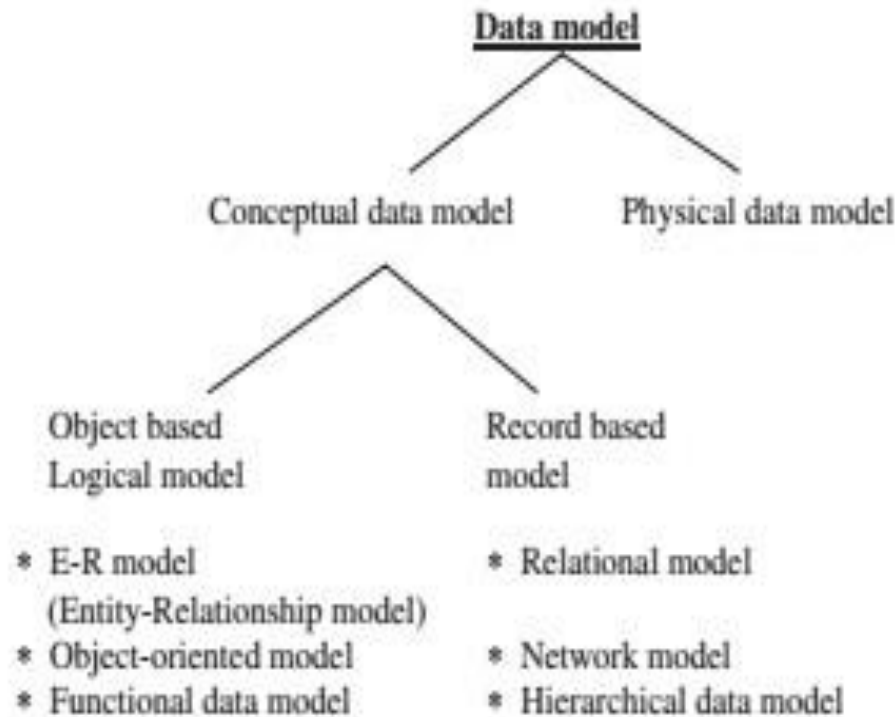
1.6 ANSI/ SPARK DATA MODEL

Example : College management System



1.7 Data Modelling

- ✓ Is the collection of conceptual tools for describing data, relationship between data and consistency constraints.
- ✓ Help in describing the structure of data at the logical level.



1.8 Components and Interfaces of DBMS

- ✓ DBMS involves five major components:
 - *Data,*
 - *Hardware,*
 - *Software,*
 - *Procedures*
 - *and users*

1.9 Database Users

- ✓ People interacting with Database:
 - Database administrator ~ Who manage the database
 - Database designer ~ Who design the application program
 - Database users

1.9 Database Users

✓ People interacting with Database:

– End Users:

- **Casual(Application programmers):** access database occasionally by sophisticated query language when needed.

(Manager)

- **Naïve:** they make up a large section of the end-user population. Learn only a few facilities that they may use repeatedly

Example: people accessing database over the web, bank tellers, . staff.

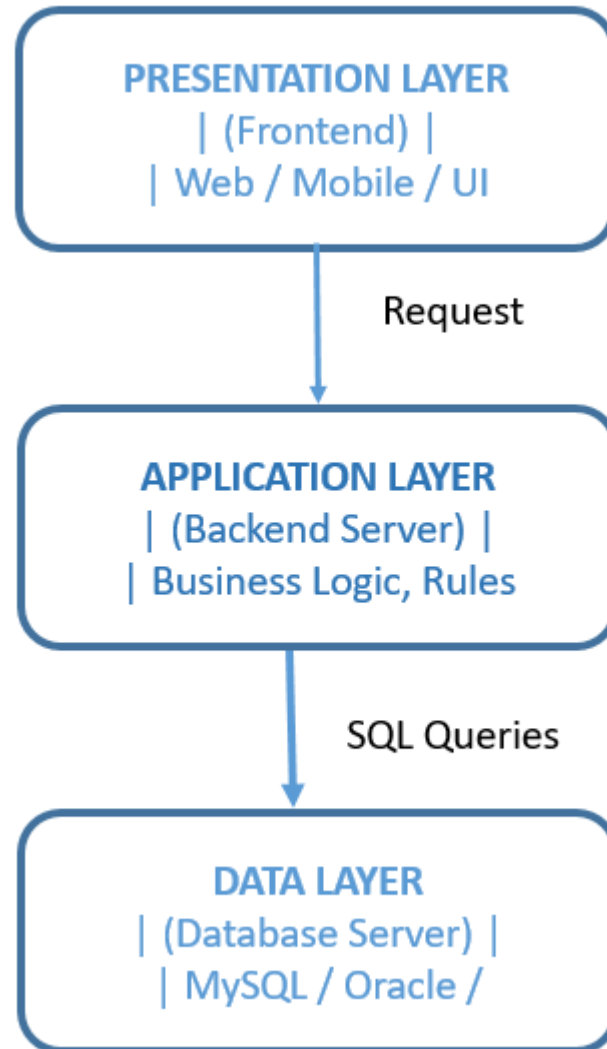
- **Sophisticated:** These include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities.

1.10 Database Architecture

Describes the location of all the pieces of information that make up the database application.

- **Frontend (presentation) layer:**
 - How the data should appear to the user and how the user should interact with the interface
- **Application logic layer:**
 - Data Processing e.g. Business Logic data validation & calculations and data manipulation.
 - Can be referred to as: Business logic/rules.
- **Data(resource) management layer:**
 - Deals with the different data sources of the Information Systems; (connecting to a database, update of data like insert, delete.).

1.10 Database Architecture

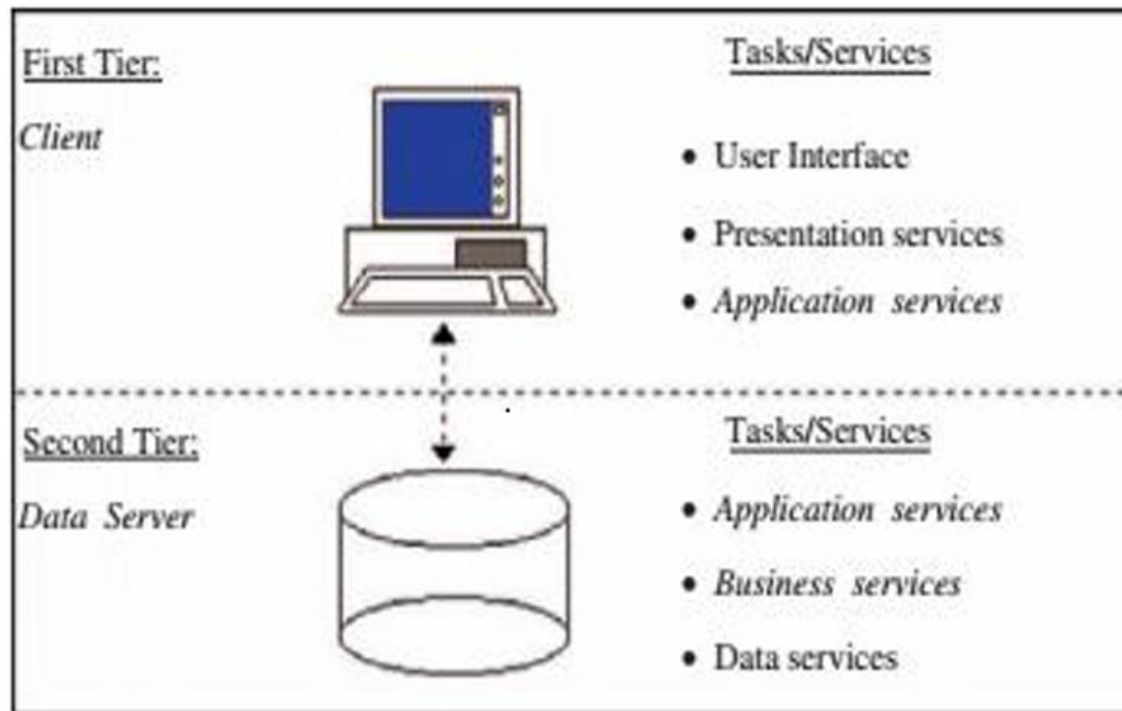


1.10 Database Architecture

✓ Describes the location of all the pieces of information that make up the database application.

1. **One tier:** user interface, application services, data running Client computer e.g. **Desktop application**

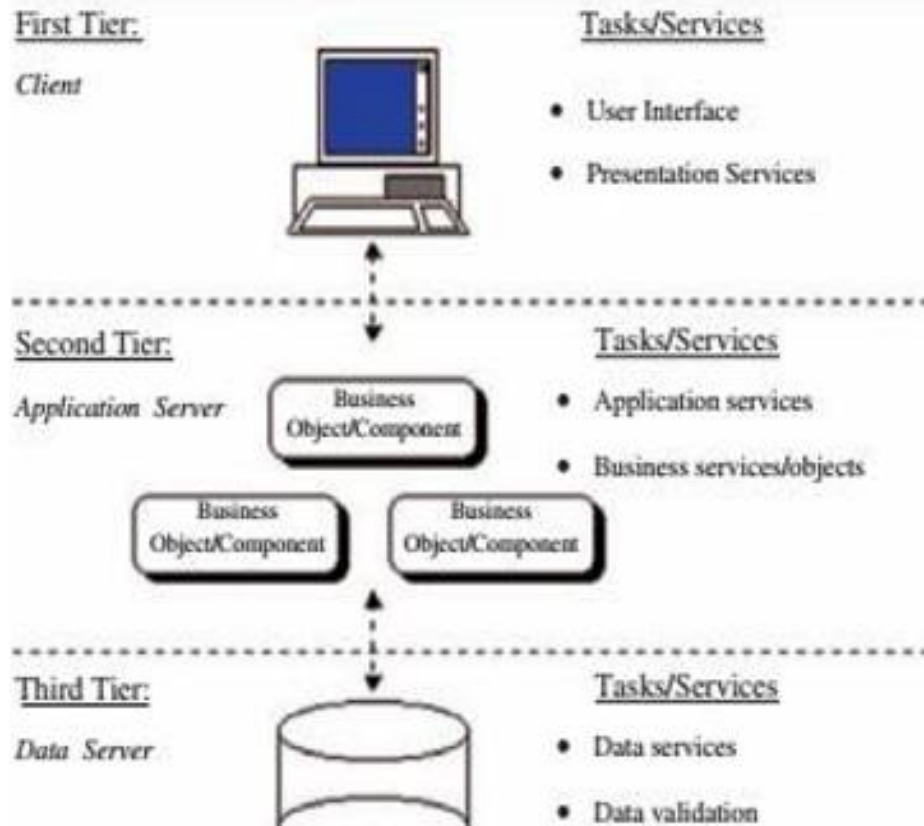
2. **Two-Tier:**



1.10 Database Architecture

- ✓ Describes the location of all the pieces of information that make up the database application

3. Three-Tier Architecture



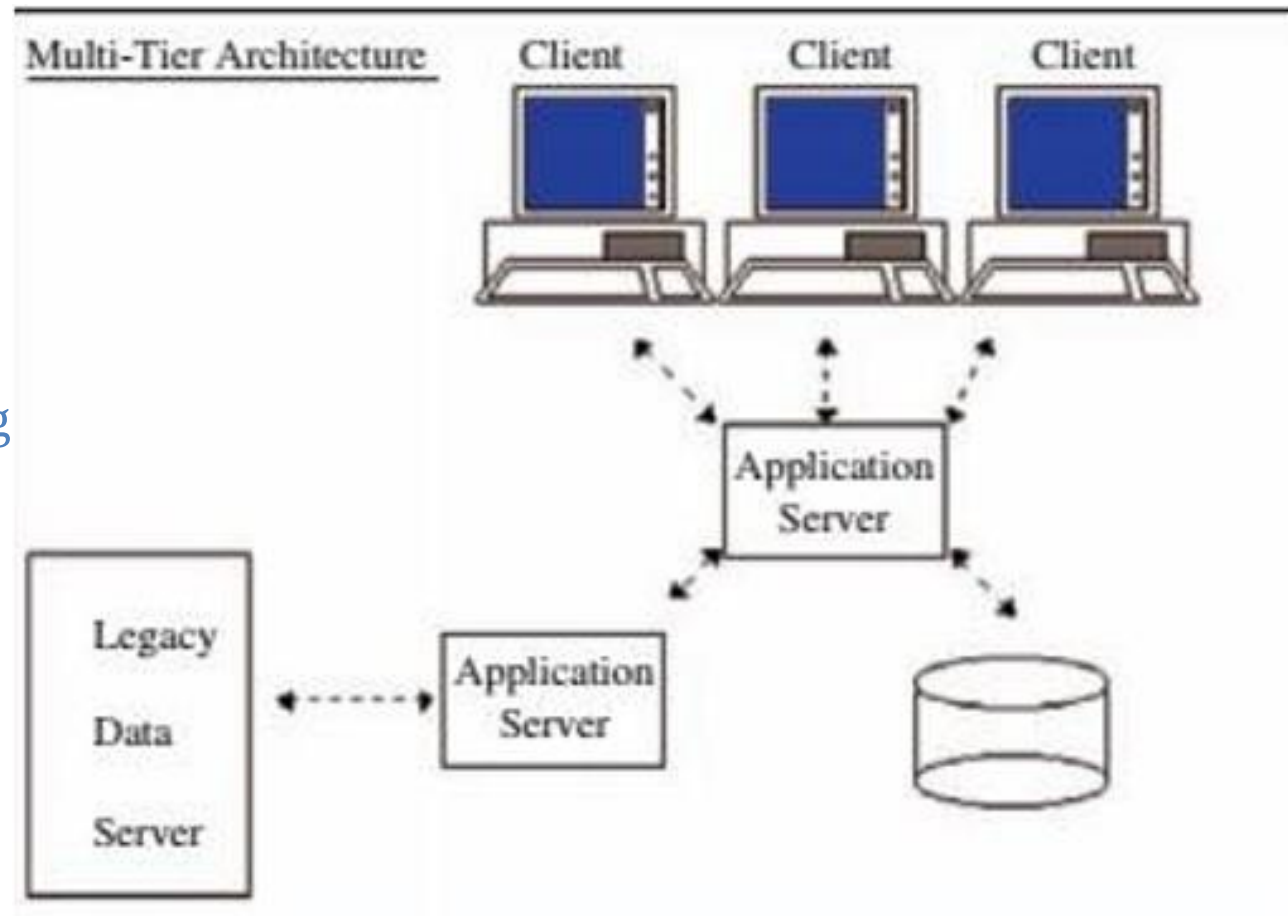
1.10 Database Architecture

- ✓ Describes the location of all the pieces of information that make up the database application

4. Multiple-Tier :

E.g: Amazon Online shopping

- User service
- Product service
- Payment service
- Order services



End

